

AKASH BAG

Software Engineer | Frontend Engineer | Full-Stack Developer
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Professional Summary

Software Engineer with 2+ years of experience building **frontend and full-stack web applications**, industrial IoT platforms, analytics dashboards, and distributed data-processing systems. Strong expertise in **React.js, Next.js, TypeScript, JavaScript, Node.js, Fastify.js, REST APIs, PostgreSQL, SQL, Docker, and Git**. Experienced in **responsive UI development, API integration, state management, role-based access control, authentication, authorization, performance optimization, asynchronous processing, and scalable system architecture**.

Professional Experience

Software Engineer

Inwork Global Pvt Ltd | Nov 2025 – Jun 2026

- Architected a distributed query orchestration system handling **500K+ records** for analytical filtering, data processing, reporting, and CSV export workflows.
- Built asynchronous worker-based processing pipelines for long-running queries, improving frontend responsiveness with **sub-second API response times** and eliminating UI blocking across report workflows.
- Optimized complex **SQL and PostgreSQL queries**, reducing analytical query execution time by **40%** through query optimization and efficient data processing.
- Developed **2+ analytics and operational dashboards** using modern frontend architecture for KPI monitoring, business intelligence, data visualization, and real-time reporting.
- Implemented **RBAC, authentication, authorization, protected routes, and permission-based UI rendering** across **4+ user roles**.
- Applied **Agentic AI workflows and AI-assisted development** to accelerate engineering delivery by **30%** while maintaining code quality, maintainability, and scalability.

Software Development Engineer L-1

IEM-UEM Group, Kolkata | Sep 2024 – Oct 2025

- Led a team of **4 engineers** to architect and deliver a university hiring portal serving **1,000+ students and recruiters**, owning system architecture, task delegation, API integration, and technical decision-making.
- Engineered a real-time **industrial IoT monitoring platform** for machine health tracking, anomaly detection, predictive maintenance, and operational analytics across **50+ connected devices**.
- Built scalable **data ingestion and processing pipelines** for high-frequency IoT telemetry with **sub-second dashboard refresh rates** and **4+ role-based monitoring dashboards**.
- Developed secure **REST API-driven workflows** for device management, telemetry processing, authentication, authorization, and role-based access control across operations and engineering teams.
- Developed reusable frontend components and responsive dashboards using **React.js, JavaScript, Tailwind CSS, API integration, and state-driven UI architecture**.

Software Developer Intern

IEMA Research & Development Pvt Ltd | Mar 2024 – Aug 2024

- Optimized a university coding competition platform serving **12,000+ students** by integrating **REST APIs** and real-time leaderboard functionality, improving data refresh latency for live event scoring.
- Developed responsive and reusable **React.js frontend components**, API-integrated interfaces, and production-ready UI features used by **500+ active users**.

Front-End Developer Intern

ReadyToGo Travels Pvt Ltd | May 2022 – Jul 2022

- Developed responsive and user-focused frontend interfaces using **HTML5, CSS3, JavaScript, and responsive design principles** while participating in code reviews and UI quality improvements.

Technical Skills

Languages: JavaScript, TypeScript, C/C++, SQL, HTML5, CSS3

Frontend: React.js, Next.js, Tailwind CSS, React Query, Responsive Design, Component Architecture, State Management, API Integration

Backend: Node.js, Fastify.js, Express.js, Payload CMS, RESTful APIs, Authentication, Authorization, RBAC, Asynchronous Processing

Databases: PostgreSQL, MySQL, MongoDB, Firebase, SQL, Database Design, Query Optimization

Tools & DevOps: Docker, Git, GitHub, CI/CD, Drizzle ORM, NextAuth

Architecture: Microservices, Distributed Systems, REST Architecture, RBAC, Scalable Systems, Event-Driven / Async Processing

Engineering Practices: Agile, Code Review, Debugging, Performance Optimization, API Design, AI-Assisted Development, Agentic AI

Education

B.Tech in Computer Science and Technology

University of Engineering & Management, Kolkata

Graduated: 2024

CGPA: 8.8 / 10

Achievements

- Architected a distributed query orchestration system handling **500K+ records**, reducing engineering delivery time by **30%** through Agentic AI workflows.
- Contributed to open-source projects across multiple **Hacktoberfest** events, earning official recognition for community contributions.
- Achieved a maximum competitive programming rating of **1,704** and solved **200+ LeetCode problems** across multiple coding competitions.

Certifications

Complete React Developer (Redux, Hooks, GraphQL) – Udemy

The Web Developer Bootcamp – Udemy

2023

2023